WORKED SOLUTION

STEP II 2025, Question 10

Mechanics • Statics of a Rigid Rod

STEP II 2025, Question 10

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

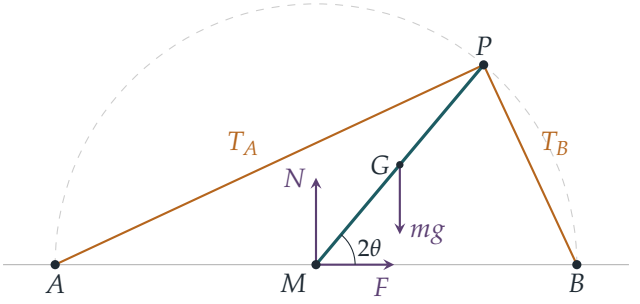
The key idea

Because $MA = MB = MP = a$, the point P lies on the circle with diameter AB, and that one observation drives everything: it gives the right angle at P, the string lengths in terms of θ, and the angles each string makes with the rod. With Hooke's law the tension is $\lambda(\text{length} - \ell)/\ell$, which splits naturally into a "$\lambda \times \text{length}/\ell$" part and a constant "$-\lambda$" part; that split is exactly the decomposition asked for in (i). After that, moments about well-chosen points remove the forces we don't want.

Solution

Model the rod as a uniform rigid rod with its weight mg acting at its midpoint G, and the strings as light elastic strings obeying Hooke's law. Put M at the origin with $B = (a, 0)$, $A = (-a, 0)$, so that

$$P = (a\cos 2\theta,\ a\sin 2\theta).$$



(The direction of the friction F in the diagram is justified in part (iii).)

Part (i)

Since $MA = MB = MP = a$, the point P lies on the circle with centre M and diameter AB. By the angle-in-a-semicircle theorem, $\angle APB = 90^\circ$.

String lengths. Triangle MPA is isosceles with $MP = MA$ and apex angle $\angle PMA = 180^\circ - 2\theta$, so its base angles are $\angle MPA = \angle MAP = \theta$. Similarly triangle MPB has apex angle 2θ, so $\angle MPB = 90^\circ - \theta$. In the right-angled triangle APB with hypotenuse $AB = 2a$ and $\angle PAB = \theta$,

$$PA = 2a\cos\theta, \qquad PB = 2a\sin\theta.$$

Since $\theta < 45^\circ$, $PB < PA$; both strings are in tension, so $2a\sin\theta > \ell$.

Splitting the force. Let $\mathbf{u}_A$ and $\mathbf{u}_B$ be the unit vectors from P towards A and B. By Hooke's law the tension in string PA is $\lambda(PA - \ell)/\ell$, so the force it exerts on the rod at P is

$$\frac{\lambda(PA-\ell)}{\ell}\,\mathbf{u}_A = \frac{\lambda}{\ell}\,PA\,\mathbf{u}_A - \lambda\mathbf{u}_A = \frac{\lambda}{\ell}\,\overrightarrow{PA} - \lambda\mathbf{u}_A,$$

and likewise for PB. Adding the two,

$$\text{total string force} = \frac{\lambda}{\ell}\left(\overrightarrow{PA} + \overrightarrow{PB}\right) - \lambda(\mathbf{u}_A + \mathbf{u}_B).$$

Now $\overrightarrow{PA} + \overrightarrow{PB} = 2\overrightarrow{PM}$ because M is the midpoint of AB, and $|\overrightarrow{PM}| = a$. So the first term is a force of magnitude $2a\lambda/\ell$ along the rod (towards M).

For the second term, $\mathbf{u}_A$ and $\mathbf{u}_B$ are perpendicular unit vectors (the angle at P is a right angle), so $\mathbf{u}_A + \mathbf{u}_B$ is the diagonal of a unit square: it has length $\sqrt{2}$ and points along the bisector of $\angle APB$. Hence $-\lambda(\mathbf{u}_A + \mathbf{u}_B)$ is a force of magnitude $\sqrt{2}\,\lambda$ along the bisector of $\angle APB$ (pointing away from AB). This is the required decomposition.

Part (ii)

Take moments about M; the friction and normal reaction there have no moment. Measure anticlockwise as positive (so B is to the right).

Ask yourself How far is M from the line of each string?

The line PA makes angle $\angle MPA = \theta$ with PM, and $PM = a$, so the perpendicular distance from M to line PA is $a\sin\theta$. Likewise the distance from M to line PB is $a\sin(90^\circ - \theta) = a\cos\theta$. The weight acts at horizontal distance $\frac{1}{2}a\cos 2\theta$ to the right of M.

The string PA pulls P towards A (anticlockwise about M); PB and the weight turn the rod clockwise. Writing T_A, T_B for the tensions,

$$T_A\,a\sin\theta \;=\; T_B\,a\cos\theta + mg\cdot\tfrac{1}{2}a\cos 2\theta.$$

With $T_A = \lambda(2a\cos\theta - \ell)/\ell$ and $T_B = \lambda(2a\sin\theta - \ell)/\ell$,

$$\begin{aligned} T_A\sin\theta - T_B\cos\theta &= \frac{\lambda}{\ell}\left(2a\cos\theta\sin\theta - \ell\sin\theta - 2a\sin\theta\cos\theta + \ell\cos\theta\right) \\ &= \lambda(\cos\theta - \sin\theta). \end{aligned}$$

(This is just the moment of the $\sqrt{2}\,\lambda$ bisector force, since the force along the rod has no moment about M.) So the moment equation becomes

$$\lambda(\cos\theta - \sin\theta) = \tfrac{1}{2}mg(\cos^2\theta - \sin^2\theta) = \tfrac{1}{2}mg(\cos\theta - \sin\theta)(\cos\theta + \sin\theta).$$

Since $0 < \theta < 45°$, $\cos\theta - \sin\theta > 0$, so we may divide by it:

$$\cos\theta + \sin\theta = \frac{2\lambda}{mg}.$$

The range of λ. Write $\cos\theta + \sin\theta = \sqrt{2}\sin(\theta + 45°)$. The rod is inclined to the ground, so $0 < \theta < 45°$, giving $45° < \theta + 45° < 90°$, where sin is strictly increasing from $\frac{1}{\sqrt{2}}$ to 1. Hence

$$1 < \cos\theta + \sin\theta < \sqrt{2} \quad\Longrightarrow\quad \tfrac{1}{2}mg < \lambda < \tfrac{1}{2}\sqrt{2}\,mg,$$

which is therefore a necessary condition for such an equilibrium.

Part (iii)

Ask yourself Which point makes both string forces disappear from a moment equation?

Both strings act at P, so take moments about P: then only the weight and the two ground forces at M appear. Relative to P, the point M is at horizontal distance $a\cos 2\theta$ (to the left) and vertical distance $a\sin 2\theta$ (below); G is at half those distances.

Suppose the friction F acts towards B (we confirm this below). About P:

- N (upwards at M, to the left of P) has clockwise moment $N\,a\cos 2\theta$;
- F (towards B at M, below P) has anticlockwise moment $F\,a\sin 2\theta$;
- mg (downwards at G, to the left of P) has anticlockwise moment $mg\cdot\frac{1}{2}a\cos 2\theta$.

For equilibrium,

$$Na\cos 2\theta = Fa\sin 2\theta + \tfrac{1}{2}mg\,a\cos 2\theta.$$

Since $2\theta < 90°$, $\cos 2\theta > 0$, and dividing by $a\cos 2\theta$ gives

$$N - F\tan 2\theta = \tfrac{1}{2}mg,$$

as required.

Direction of friction. Resolving horizontally, F balances the horizontal part of the string force. From (i), $\mathbf{u}_A = (-\cos\theta, -\sin\theta)$ and $\mathbf{u}_B = (\sin\theta, -\cos\theta)$,

and $\overrightarrow{PM} = (-a\cos 2\theta, -a\sin 2\theta)$, so the string force has horizontal component $-\frac{2a\lambda}{\ell}\cos 2\theta + \lambda(\cos\theta - \sin\theta)$. Hence the friction towards B is

$$F = \frac{2a\lambda}{\ell}\cos 2\theta - \lambda(\cos\theta - \sin\theta) = \lambda(\cos\theta - \sin\theta)\left(\frac{2a(\cos\theta + \sin\theta)}{\ell} - 1\right),$$

which is positive because $2a\sin\theta > \ell$ (both strings taut). So friction does act towards B, and F is its magnitude.

Checks

- The limits in (ii) make sense: as $\theta \to 0$ the rod lies nearly flat and $\lambda \to \frac{1}{2}mg$; as $\theta \to 45^\circ$ the rod becomes vertical and $\lambda \to mg/\sqrt{2}$.
- Numerically, with $a = 1$, $mg = 10$, $\theta = 20^\circ$, $\ell = 0.5$: then $\lambda = 5(\cos 20^\circ + \sin 20^\circ) \approx 6.408$, the tensions are $T_A \approx 17.68$ and $T_B \approx 2.36$, and resolving gives $N \approx 18.26$, $F \approx 15.81$, so $N - F\tan 40^\circ \approx 5.00 = \frac{1}{2}mg$. Moments about M also balance. A computer check over thousands of random configurations confirmed every result to rounding error.

Common slips

- Taking the tension as $\lambda \times$ extension rather than $\lambda \times$ extension$/\ell$.
- Dividing by $\cos\theta - \sin\theta$ or $\cos 2\theta$ without noting that $\theta < 45^\circ$ makes them positive.
- Taking moments about M in (iii): it removes N and F, which are exactly the forces we want. Moments about P remove the strings instead.

Beyond STEP

The split in (i) is a neat instance of a general fact: a set of forces acting on a rigid body is equivalent to a single force plus a couple, and you are free to group forces however makes the moments easiest. The "$-\lambda$ per string" piece here behaves like a pair of constant-tension strings, whose resultant is fixed by geometry alone. Statics of rigid bodies, virtual work and stability of equilibria are developed in Kleppner and Kolenkow, *An Introduction to Mechanics*.

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